JUNIOR DATA SCIENTIST PORTFOLIO

Retail Sales Forecasting

Predicting Future Sales Demand from Historical Retail Transactions

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540,000+

Real Transactions

UCI Machine Learning Dataset
with 1-year historical UK retail orders

19.84% MAPE

Forecast Accuracy

Random Forest Model outperforms
Holt-Winters ETS & Seasonal Baseline

£10.64 Million

Gross Revenue Modeled

Actionable inventory safety buffer
recommendations (15-20%) for ops

SWIPE TO EXPLORE PIPELINE >>>

STEP 1 — DATA PREPROCESSING

Data Cleaning & Business Rules

Why domain context matters more than blindly dropping records

Data Audit & Funnel

Raw Dataset Rows	**541,909**	100%
Identical Duplicates Removed	**5,268**	0.97%
Cancelled Invoices ('C') Filtered	**9,251**	1.71%
Warehouse Adjustments (Price <= 0)	**2,512**	0.46%
Clean Valid Transactions	**524,878**	96.86%

Key Business-Driven Decisions

1. Preserved 132k Missing CustomerIDs (~25%)

Guest checkouts still generate real store revenue. Dropping anonymous buyers would underestimate true demand by ~25%.

2. Filtered Price <= £0 Records (2,512 rows)

Descriptions like 'damaged', 'lost', 'check' prove these were internal warehouse notes, NOT consumer sales transactions.

3. Handled Cancellations & Negative Quantities

Cancelled invoices ('C') and negative quantities separated to accurately capture Gross Sales Demand without skew.

4. Feature Engineering: TotalAmount

Computed as Quantity x UnitPrice to serve as our daily target.

STEP 2 — EXPLORATORY DATA ANALYSIS

Uncovering Operational Cycles & Trends

Visualizing 1-year historical demand behavior

Day of Week Sales Distribution

Critical Operational Findings

- **Saturday Zero-Order Phenomenon**
  The online store registered £0 sales on every Saturday throughout the year (warehouse and order system closed).
- **Mid-Week Peak Demand**
  Thursday (£2.20M) and Tuesday (£2.18M) drive the highest volume of retail purchases across the entire year.
- **Q4 Year-End Surge**
  Sales velocity accelerates rapidly starting September into November due to early holiday and Christmas gifting.
- **Incomplete December Data Anomaly**
  Data cuts off on Dec 9. The drop in monthly revenue is an artifact of partial data, NOT business decline.

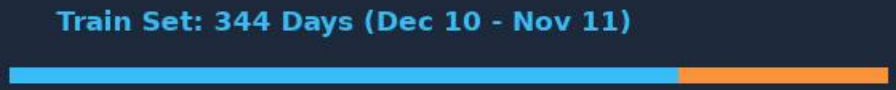
STEP 3 — FORECASTING STRATEGY

Why Baseline Models Are Essential

Validating whether machine learning is truly necessary

Time-Based Train/Test Split

Strict chronological ordering avoids data leakage:



Evaluation Horizon: 30 Days (Peak Q4)

- Validation: 2011-11-12 to 2011-12-11
- High volatility period (Black Friday & early Christmas)
- Strict out-of-sample evaluation

Baseline Comparison Insights

Baseline 1: Standard Naive (Yesterday) MAE: 619,212 | MAPE: 42.76%

Predicting today = yesterday fails catastrophically when transitioning from Friday into Saturday (closed), and Sunday into Monday.

Baseline 2: Seasonal Naive (7-Day Lag) MAE: 338,178 | MAPE: 24.27%

Predicting today = same day last week cuts error nearly in half (-45%) by naturally mimicking the 7-day operational work cycle!

Core Data Science Takeaway Seasonal Baseline > Complex Naive

A model that understands calendar seasonality beats an agnostic predictor before any machine learning algorithm is even applied.

STEP 4 — MODEL BENCHMARKING

Model Evaluation on 30-Day Test Horizon

Comparing Baselines, Statistical Holt-Winters ETS, and Random Forest

Actual vs Predicted Sales Performance

Evaluation Scoreboard

Model	MAE (£)	MAPE
Random Forest (ML)	**£14,413**	**19.84%**
Seasonal Naive 7-Day	£15,372	24.27%
Holt-Winters ETS	£16,895	24.14%
7-Day Moving Avg	£24,615	31.78%
Standard Naive	£29,825	42.76%

Why Random Forest Won:
Combined multi-period lag momentum (Lag 1, 2) with 7-day cyclical seasonality and calendar signals.

STEP 5 — ACTIONABLE RECOMMENDATIONS

Translating Forecasting into Business Impact

How retail stakeholders can use these numbers in practice

1. Inventory Safety Buffer (15% - 20%)

With a mean absolute daily error of ~£14.4k during peak Q4:

- Fast-moving products require a safety buffer of 15-20% to mitigate stockout risk without inflating holding costs.
- Avoid ordering excess inventory for Saturday arrivals.

2. Warehouse Staffing Optimization

Align workforce schedules with proven demand peaks:

- Schedule peak packing and dispatch shifts on Tue & Thu.
- Zero packing staff needed on Saturdays (zero orders).
- Reduce Sunday operations to half-capacity.

3, Project Limitations & Enterprise Roadmap

- 1-Year Horizon Limitation: Dataset spans only 12 months; multi-year cyclical patterns cannot be verified.
- Missing Exogenous Features: Future iterations should ingest promotion calendars, ad spend, and SKU-level data.
- Full Reproducible Pipeline: Available on GitHub with interactive Google Colab notebook support.